

Justify your answers and show your work. Unless stated otherwise, all questions worth 5pts.

1. [2 pts each] Answer each question.

a. Write the infinite series, $2 + 4 + 6 + 8 \cdots$, using sigma notation (i.e. using Σ).

$$\sum_{k=1}^{\infty} 2k$$

b. Evaluate the series $\sum_{k=2}^n (2k + 1)$ for $n = 4$.

$$\sum_{k=2}^4 (2k + 1) = 5 + 7 + 9 = 21$$

c. If $0 < a_n < b_n$ and $\{a_n\}$ diverges then what can be said about $\{b_n\}$?

$a_n < b_n \Rightarrow b_n$ must diverge.

d. If $\lim_{n \rightarrow \infty} \{a_n\} = 0$ then what does the Divergence Test say about $\sum_{k=0}^{\infty} a_k$?

The test is inconclusive.

e. When approximating $\sqrt[3]{27.5}$ with a Taylor polynomial, what function and center should be used.

$f(x) = \sqrt[3]{x}$ and $a = 27$.

Evaluate the following sequences.

2. $\lim_{n \rightarrow \infty} \left\{ \frac{n^3 - 2n^2}{5n^3 + 2n} \right\}$

$$\lim_{n \rightarrow \infty} \left\{ \frac{n^3 - 2n^2}{5n^3 + 2n} \right\} = \lim_{n \rightarrow \infty} \left\{ \frac{n^3/n^3 - 2n^2/n^3}{5n^3/n^3 + 2n/n^3} \right\} = \lim_{n \rightarrow \infty} \left\{ \frac{1}{5} \right\} = \frac{1}{5}$$

3. $\lim_{n \rightarrow \infty} \left\{ \frac{0.3^n}{n^{10}} \right\}$

$n^{10} \ll 0.3^n \Rightarrow$ the sequence diverges.

Evaluate the geometric series or state that it diverges.

4. $\sum_{k=0}^{\infty} 5 \frac{4^k}{3^k}$

$$\sum_{k=0}^{\infty} 5 \frac{4^k}{3^k} = 5 \sum_{k=0}^{\infty} \left(\frac{4}{3} \right)^k \Rightarrow r = \frac{4}{3} > 1 \text{ so the series diverges.}$$

5. $\sum_{k=2}^{\infty} 3 \left(\frac{1}{2} \right)^k$

$$3 \sum_{k=2}^{\infty} \left(\frac{1}{2} \right)^k = 3 \sum_{k=0}^{\infty} \left(\frac{1}{2} \right)^{k-2} = \frac{3}{4} \sum_{k=0}^{\infty} \left(\frac{1}{2} \right)^k = \frac{3}{4} \frac{1}{1 - 1/2} = \frac{6}{4} = \frac{3}{2} \text{ since } r = 1/2 < 1.$$

Use the Divergence Test to determine whether the following series diverge, or state that the test is inconclusive.

6. $\sum_{k=1}^{\infty} \frac{k^2 - 1}{3k^2 + 10}$

$$\lim_{k \rightarrow \infty} \left\{ \frac{k^2 - 1}{3k^2 + 10} \right\} = \frac{1}{3} \text{ (similarly to number 2) which is not 0 so it diverges.}$$

Using the Squeeze Theorem, find the limit of the following sequence or determine that the limit does not exist.

7. $\lim_{n \rightarrow \infty} \left\{ \frac{(-1)^n}{n} \right\}.$

$$0 = \left\{ \frac{-1}{n} \right\}_{n=1}^{\infty} \leq \left\{ \frac{(-1)^n}{n} \right\}_{n=1}^{\infty} \leq \left\{ \frac{1}{n} \right\}_{n=1}^{\infty} = 0 \text{ so then by the Squeeze Theorem, } \lim_{n \rightarrow \infty} \left\{ \frac{(-1)^n}{n} \right\} = 0.$$

Use the Integral Test to determine whether the series converges, diverges, or state why the test cannot be used. Check that the conditions of the test are satisfied.

8. [10 pts] $\sum_{k=1}^{\infty} \frac{1}{x^{2/3}}.$

Let $f(x) = x^{-2/3}$. $f(x) > 0$ and $f'(x) = -\frac{2}{3}x^{-5/3} < 0$ for $x \geq 1$. So then we can use the integral test.

$$\int_1^{\infty} x^{-2/3} dx = \lim_{b \rightarrow \infty} 3x^{1/3} \Big|_1^b = \infty - 3 = \infty.$$

Use the function $f(x) = \sqrt{x}$ to answer the following.

9. Find the Taylor polynomial of order $n = 3$ for $f(x)$ centered at 1.

$$f'(x) = \frac{1}{2}x^{-1/2}, f''(x) = -\frac{1}{4}x^{-3/2}, \text{ and } f'''(x) = \frac{3}{8}x^{-5/2}. \text{ So then when } a = 1,$$

$$p_3(x) = 1 + \frac{1}{2}(x-1) - \frac{1}{8}(x-1)^2 + \frac{1}{16}(x-1)^3$$

10. Use the Taylor polynomial from above to approximate $\sqrt{1.05}$.

$$p_3(1.05) = 1.024695$$

11. Compute the absolute error in the approximation assuming the exact value is given by a calculator.

$$\text{error} = \left| p_3(1.05) - \sqrt{1.05} \right| \approx 0.0000000235$$